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HW #5

McCloskey
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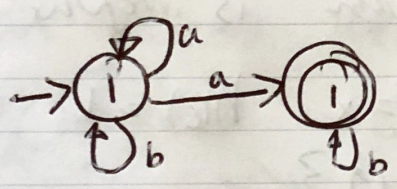
1) a)

	a	b	init	accep
P_0Q_0	P_1Q_1	P_2Q_0	✓	✓
P_2Q_0	P_2Q_1	P_0Q_0	×	✓
P_1Q_1	P_0Q_1	P_2Q_0	×	·
P_0Q_1	P_1Q_1	P_2Q_0	·	×
P_2Q_1	P_2Q_1	P_0Q_1	×	·

b) change the accepting states such that the state contains at LEAST one P_0, P_1, Q_0

c) change the accepting states such that the state contains at least one P_0/P_1 and Q_1 (P_0Q_1) etc

2)



accepts the state $a((1))$ no matter what

5) (a) given the DFA to find L, one would append the DFA to find the string x to the beginning of the DFA of L to find the prefix of x

(b) similar to that of part (a), you append the DFA to find x to the end of L, giving you the suffix containing x, and create an accepting state after

(c) given the language to find L, create a DFA accepting L, and also create a DFA accepting the occurence of x, and combine the two DFA ~~to create~~ by intersection and create accepting states that show intersection

6) (a) $\{a^i b^j c^k \mid i+j=k\}$ – assume regular

Proof by contradiction

$i=0 \Rightarrow a^0 b^j c^k \Rightarrow b^j c^k \in L \Rightarrow b^i c^i$

but,

$b^i c^i \Rightarrow bbbb \ldots b^i c^i \boxed{\notin L}$

also

$b^i c^i \Rightarrow b^i c^i cccc \ldots c \boxed{\notin L}$

lastly

$b^i c^i \Rightarrow bcbcbc \ldots b^i c^i \boxed{\notin L}$

not regular

because they are not contained in the regular language (L), the language is irregular

(b) $h(a)=0$ $h(b)=x$ $h(c)=1$

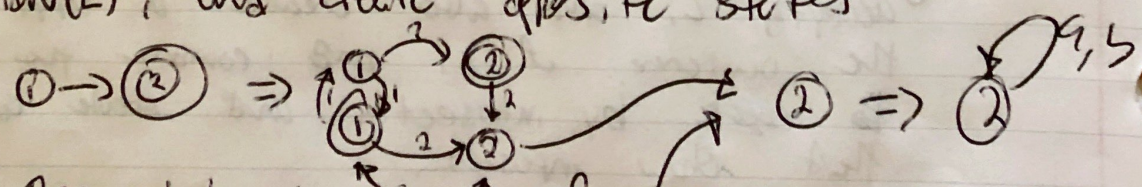
$h(L) = \{0^i x^i 1^{i+j}\}$

if $j=0$

$h(L) = \{0^i 1^i\}$ which is the ideal irregular language

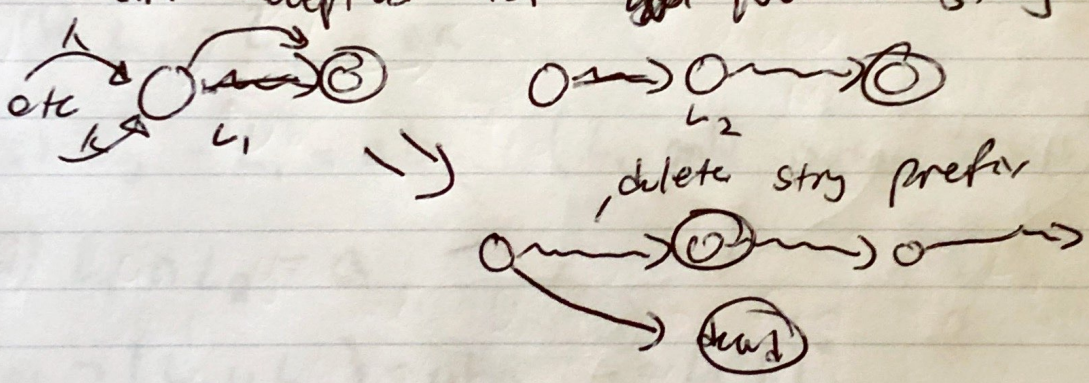
L is not regular

3) The DFA for L to be changed into the DFA for MIN(L) requires you to create the initial states, which are the accepting states for L. Secondly create identical states in the MIN(L), and create opposite states



These duplicated states, if they are non-accepting make them dead

4) (a) to create the $x \backslash L$ DFA from the DFA for L, x must be a prefix. if this does not occur, create the dead state from x. to do this the transitions must be parallel or series

because the DFA for the new one can be constructed the DFA for x is a regular language

(b) to find the DFA for $L_1 \backslash L_2$ we need to find each prefix that can be created from L_1, and transition to each one using λ-transitions. append this to the L_2 DFA, given that when L_1 is found in L_2, it is removed from the string, giving an accepted left quotient string